Changing velocity

antocciphysics | TEACHER KEY | CPA + MATCHING SEI

Learning target: Calculate acceleration and distinguish speeding up, slowing down, and turning.

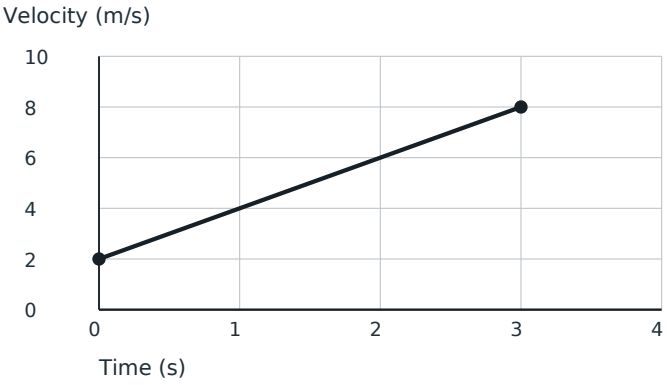
Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: read motion in two ways *Review: Unit 1, Day 4.*
(a) 4 × 6 = 24 m west (displacement -24 m if east is positive). (b) Draw an 8-unit east component and a 3-unit south component; the resultant points southeast, closer to east than south.

02. Worked example: braking on a bike
Both velocity arrows point east. The final arrow is one-quarter the initial length (2/8). The separate acceleration arrow points west. Acceleration opposes velocity while the cyclist slows; she does not reverse direction.

03. A remote-control car speeds up
(a) +2 m/s and +8 m/s. (b) Δv = 8 - 2 = +6 m/s; a = 6/3 = +2 m/s², east. (c) Velocity stays positive and increases, so its magnitude (speed) increases. An upward slope by itself does not always mean speeding up.

Model / diagram check



04. Slowing before a crosswalk
(a) Speed lost = 10 - 4 = 6 m/s over 5 s, or 1.2 m/s each second. (b) With east positive, a = (4 - 10)/5 = -1.2 m/s²: 1.2 m/s² west. She is still moving east at 4 m/s.

05. Running around a bend
Disagree. Draw equal-length arrows tangent to the track at the two points. Their directions differ, so velocity changes even though speed stays the same. A changing velocity means acceleration.

06. Review: a robot's route *Review: Unit 1, Day 6.*
East-west change: 5 - 2 = 3 m east. North-south change: 3 - 7 = -4 m, or 4 m south. The route must join all four moves in order; the endpoint is southeast of the start.

07. Exit check: a docking simulation
Slowing down. With east positive, v_i = -6 m/s and v_f = -2 m/s. a = [-2 - (-6)]/2 = +2 m/s², east. The acceleration opposes the westward velocity, reducing the speed.

08. STRETCH: negative can mean faster
A straight line joins (0 s, -1 m/s) to (2 s, -5 m/s). a = [-5 - (-1)]/2 = -2 m/s², west. Both velocity and acceleration point west; the speed increases from 1 to 5 m/s.

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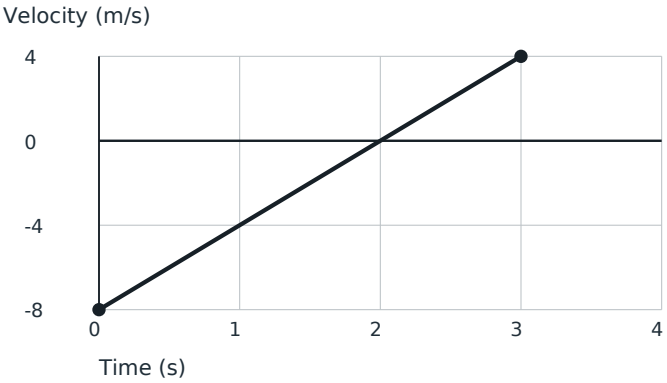
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01. Warm-up: motion and direction *Review: Unit 1, Day 4.*
(a) Legs: 24 m east and 12 m west. Distance = 24 + 12 = 36 m; displacement = 24 - 12 = 12 m east. (b) The reference direction or axis and which way the angle is measured, such as 30° north of east.

02. Worked example: braking on a bike
Both velocity arrows point east. The final arrow is one-quarter the initial length (2/8). The separate acceleration arrow points west. Acceleration opposes velocity while the cyclist slows; she does not reverse direction.

03. A test car changes direction
(a) $a = [4 - (-8)]/3 = +4$ m/s², east. (b) $v = -8 + 4t = 0$ at $t = 2$ s. (c) From 0 to 2 s it moves west and slows. From 2 to 3 s it moves east and speeds up. It reverses at 2 s; acceleration remains east throughout.

Model / diagram check



04. Braking for stopped traffic
(a) $a = (0 - 24)/7.0 = -3.43$ m/s² (about -3.4 m/s²). (b) Average velocity = (24 + 0)/2 = 12 m/s; displacement = 12 × 7.0 = +84 m. (c) Acceleration points backward while the car continues moving forward. This is braking distance only, excluding driver reaction time.

05. Can a tiny thruster matter?
(a) Shade the rectangle from $t = 0$ to 1×10^6 s, between $a = 0$ and 2×10^{-6} m/s². Its area is $\Delta v = (2 \times 10^{-6}\ \text{m/s}^2)(1 \times 10^6\ \text{s}) = 2$ m/s in the thrust direction. (b) The duration matters: a small acceleration can act for a long time. Whether 2 m/s matters to the mission depends on its requirements.

06. Review: return the robot to its start *Review: Unit 1, Day 6.*
(a-b) With east +x and north +y, A = (5, -3) m and B = (-3, 7) m. R = (2, 4) m; $|R| = \sqrt{(2^2 + 4^2)} = \sqrt{20} \approx 4.47$ m, 63.4° north of east. (c) Return move C = (-2, -4) m: 2 m west and 4 m south. Then A + B + C = 0.

07. Exit check: thrusters on, then off
First interval: $a = [-9 - (-3)]/4 = -6/4 = -1.5$ m/s², west. It moves west and speeds up from 3 to 9 m/s. Second interval: $a = 0$ m/s². It continues west at a constant speed of 9 m/s.

08. STRETCH: past the turning point
(a) $v = -6 + 2t$: a straight line from (0 s, -6 m/s) to (5 s, +4 m/s). It reverses at 3 s. (b) Signed graph areas are $-\frac{1}{2}(3)(6) = -9$ m and $+\frac{1}{2}(2)(4) = +4$ m. Displacement = -9 + 4 = -5 m (5 m west); distance = 9 + 4 = 13 m. Distance counts both legs positively; displacement tracks the net change in position.